

Lecture 2: Log linearization

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Business cycle is fluctuation / deviation around trend. log linearization: take logs to linearize the system and then take deviation from the trend / steady state.

* Can not work with incontinuous functions. eg: zero lower bound, default, work choice ~

6 Taylor expansions.

Taylor expansion of order k of function f around a is to derive the best approximation of f around a by a polynomial of order k .

Suppose f is $n+1$ times differentiable in an interval $X \in (a-\varepsilon, a+\varepsilon)$ (around a).

$$f(x) = f(a) + \sum_{i=1}^n \frac{f^{(i)}(a)}{i!} (x-a)^i + \underbrace{o[(x-a)^n]}_{\text{Peano remainder terms.}}$$

eg: 1st order : $f(x) \approx f(a) + \frac{f'(a)}{1!} (x-a)^1 = f(a) + f'(a)(x-a)$

2nd order : $f(x) \approx f(a) + \frac{f'(a)}{1!} (x-a)^1 + \frac{f''(a)}{2!} (x-a)^2 = f(a) + f'(a)(x-a) + \frac{f''(a)}{2} (x-a)^2$

Taylor expansion on 2 dimensions.

Suppose f is second-order continuous (C^2) in a circle around (x_0, y_0) that contains (x_0+h, y_0+k) : For a small move (h, k) around (x_0, y_0) how to approximate changes in f .

$$f(x_0+h, y_0+k) = f(x_0, y_0) + \boxed{f_1(x_0, y_0)h + f_2(x_0, y_0)k} \text{ linear part (gradient step)} \\ + \frac{1}{2} \boxed{[f_{11}(x, y)h^2 + 2f_{12}(x, y)hk + f_{22}(x, y)k^2]} \text{ quadratic correction with Hessian}$$

where $(x, y) = (x_0 + ch, y_0 + ck)$ for some $c \in (0, 1)$

A vector format: $f(z_0 + \Delta) = f(z_0) + \nabla f(z_0)^T \Delta + \frac{1}{2} \Delta^T Hf(z_0 + c\Delta) \Delta$
where $z_0 = (x_0, y_0)$, $\Delta = (h, k)$

6 log linearization: $\hat{x}_t = \log X_t - \underbrace{\log \bar{X}}_{\text{steady state}} \rightarrow \text{percentage deviation}$

when γ is sufficiently small

since $\gamma_t \rightarrow 0$, we have

$$e^{\gamma_t} = e^0 + e^0(\gamma_t - 0) = 1 + \gamma_t \Rightarrow \gamma_t \approx \log(1 + \gamma_t)$$

$$\hat{x}_t = \log \frac{X_t}{\bar{X}} = \log \left(1 + \frac{X_t - \bar{X}}{\bar{X}} \right) \approx \frac{X_t - \bar{X}}{\bar{X}}$$

$\hat{x}_t \times 100\%$ is approximately the percentage deviation w.r.t. S.S.

$$X_t = \bar{X} \left(\frac{X_t}{\bar{X}} \right) = \bar{X} e^{\log \frac{X_t}{\bar{X}}} = \bar{X} e^{\hat{x}_t}$$

(10)

Taking first-order Taylor approximation:

$$\bar{X} e^{\hat{x}_t} \approx \bar{X} e^0 + \bar{X} e^0 (\hat{x}_t - 0) = \bar{X} + \bar{X} \hat{x}_t = (1 + \hat{x}_t) \bar{X}$$

$$\Rightarrow X_t = \underbrace{\bar{X}}_{\text{Steady state}} + \underbrace{\bar{X} \hat{x}_t}_{\text{deviation from steady state}}$$

Examples:

$$\begin{aligned} \textcircled{1} \quad Y_t &= a X_t \\ X_t &\approx \bar{X} (1 + \hat{x}_t) \end{aligned} \quad \left| \begin{aligned} \Rightarrow Y_t &\approx a \bar{X} (1 + \hat{x}_t) = a \bar{X} + a \bar{X} \hat{x}_t \\ Y_t &= a X_t \quad \forall t \Rightarrow \bar{Y} = a \bar{X} \end{aligned} \right| \Rightarrow Y_t \approx \bar{Y} + \bar{Y} \hat{y}_t, \quad Y_t \approx \bar{Y} + \bar{Y} \hat{y}_t$$

$$\Rightarrow \hat{x}_t = \hat{y}_t$$

$$\begin{aligned} \textcircled{2} \quad Z_t &= X_t Y_t \\ X_t &\approx \bar{X} (1 + \hat{x}_t) \\ Y_t &\approx \bar{Y} (1 + \hat{y}_t) \end{aligned} \quad \left| \begin{aligned} \Rightarrow Z_t &\approx \bar{X} \bar{Y} (1 + \hat{x}_t)(1 + \hat{y}_t) = \bar{X} \bar{Y} (1 + \hat{x}_t + \hat{y}_t + \hat{x}_t \hat{y}_t) \\ \Rightarrow Z_t &\approx \bar{X} \bar{Y} (1 + \hat{x}_t + \hat{y}_t) \\ Z_t &= X_t Y_t, \forall t \Rightarrow \bar{Z} = \bar{X} \bar{Y} \end{aligned} \right| \Rightarrow Z_t \approx \bar{Z} + \bar{Z} (\hat{x}_t + \hat{y}_t), \quad Z_t \approx \bar{Z} + \bar{Z} \hat{z}_t$$

$$\Rightarrow \hat{z}_t = \hat{x}_t + \hat{y}_t$$

$$\textcircled{3} \text{ Log linearize equation } X_t Y_t = C \Rightarrow \bar{X} \bar{Y} = C$$

$$\text{by } \textcircled{2} \Rightarrow X_t Y_t \approx \bar{X} \bar{Y} (1 + \hat{x}_t + \hat{y}_t) \Rightarrow C \approx C (1 + \hat{x}_t + \hat{y}_t) \Rightarrow \hat{x}_t + \hat{y}_t = 0$$

$$\textcircled{4} \quad \frac{X_t}{Y_t} = C \Rightarrow \frac{\bar{X}}{\bar{Y}} = C$$

$$\begin{aligned} \frac{X_t}{Y_t} &\approx \frac{\bar{X} (1 + \hat{x}_t)}{\bar{Y} (1 + \hat{y}_t)} \\ (1 + \hat{y}_t)(1 - \hat{y}_t) &= 1 + \hat{y}_t - \hat{y}_t - \hat{y}_t^2 \approx 1 - \hat{y}_t^2 \approx 1 \end{aligned} \quad \left| \Rightarrow \frac{X_t}{Y_t} = \frac{\bar{X}}{\bar{Y}} (1 + \hat{x}_t)(1 - \hat{y}_t) \right.$$

$$\Rightarrow \frac{X_t}{Y_t} = \frac{\bar{X}}{\bar{Y}} (1 + \hat{x}_t - \hat{y}_t - \hat{x}_t \hat{y}_t) = \frac{\bar{X}}{\bar{Y}} (1 + \hat{x}_t - \hat{y}_t) \quad \left| \Rightarrow C = C (1 + \hat{x}_t - \hat{y}_t) \right.$$

$$\Rightarrow \hat{x}_t - \hat{y}_t = 0$$

$$\textcircled{5} \quad X_t + Y_t = C \Rightarrow \bar{X} + \bar{Y} = C$$

$$X_t + Y_t = \bar{X} + \bar{Y} = C$$

$$X_t + Y_t \approx \bar{X} (1 + \hat{x}_t) + \bar{Y} (1 + \hat{y}_t) = \bar{X} + \bar{Y} + \bar{X} \hat{x}_t + \bar{Y} \hat{y}_t \Rightarrow \bar{X} \hat{x}_t + \bar{Y} \hat{y}_t = 0$$

⑥ budget constraint: $C_t + K_t = Y_t + (1-\delta)K_{t-1}$

$\Rightarrow \bar{C} + \bar{K} = \bar{Y} + (1-\delta)\bar{K}$

$C_t \approx \bar{C}(1 + \hat{c}_t), K_t \approx \bar{K}(1 + \hat{k}_t), Y_t \approx \bar{Y}(1 + \hat{y}_t), K_{t-1} \approx \bar{K}(1 + \hat{k}_{t-1})$

$\Rightarrow \bar{C}(1 + \hat{c}_t) + \bar{K}(1 + \hat{k}_t) = \bar{Y}(1 + \hat{y}_t) + (1-\delta)\bar{K}(1 + \hat{k}_{t-1})$

$\Rightarrow \bar{C} + \bar{C}\hat{c}_t + \bar{K} + \bar{K}\hat{k}_t = \bar{Y} + \bar{Y}\hat{y}_t + (1-\delta)\bar{K} + (1-\delta)\bar{K}\hat{k}_{t-1} \quad \Bigg| \Rightarrow$
 $\bar{C} + \bar{K} = \bar{Y} + (1-\delta)\bar{K}$

$\bar{C}\hat{c}_t + \bar{K}\hat{k}_t = \bar{Y}\hat{y}_t + (1-\delta)\bar{K}\hat{k}_{t-1}$

⑦ Utility FOC: $C_t^{-\bar{\epsilon}} = \lambda_t \Rightarrow \bar{C}^{-\bar{\epsilon}} = \bar{\lambda}$

$C_t = \bar{C}e^{\hat{c}_t}, \lambda_t = \bar{\lambda}e^{\hat{\lambda}_t}$

$\Rightarrow C_t^{-\bar{\epsilon}} = \bar{C}^{-\bar{\epsilon}} e^{-\bar{\epsilon}\hat{c}_t} = \bar{\lambda}e^{\hat{\lambda}_t} \quad \Bigg| \Rightarrow \begin{cases} e^{-\bar{\epsilon}\hat{c}_t} = e^{\hat{\lambda}_t} \\ e^{-\bar{\epsilon}\hat{c}_t} \approx 1 - \bar{\epsilon}\hat{c}_t \\ e^{\hat{\lambda}_t} \approx 1 + \hat{\lambda}_t \end{cases} \Rightarrow \begin{cases} 1 - \bar{\epsilon}\hat{c}_t = 1 + \hat{\lambda}_t \\ \Rightarrow -\bar{\epsilon}\hat{c}_t = \hat{\lambda}_t \end{cases}$

⑧ C-D production function: $Y_t = A_t K_t^\alpha N_t^{1-\alpha} \Rightarrow \bar{Y} = \bar{A} \bar{K}^\alpha \bar{N}^{1-\alpha}$

$\log Y_t = \log A_t + \alpha \log K_t + (1-\alpha) \log N_t$

$\log \bar{Y} = \log \bar{A} + \alpha \log \bar{K} + (1-\alpha) \log \bar{N} \quad \Bigg| \Rightarrow$

$(\log Y_t - \log \bar{Y}) = (\log A_t - \log \bar{A}) + \alpha (\log K_t - \log \bar{K}) + (1-\alpha) (\log N_t - \log \bar{N})$

$\Rightarrow \hat{y}_t = \hat{a}_t + \alpha \hat{k}_t + (1-\alpha) \hat{n}_t$

Or, directly: $\bar{Y}(1 + \hat{y}_t) = \bar{A}(1 + \hat{a}_t) \bar{K}^\alpha (1 + \hat{k}_t)^\alpha \bar{N}^{1-\alpha} (1 + \hat{n}_t)^{1-\alpha}$

$\Rightarrow 1 + \hat{y}_t = (1 + \hat{a}_t) (1 + \hat{k}_t)^\alpha (1 + \hat{n}_t)^{1-\alpha}$ let $g(\epsilon) = (1 + \epsilon)^\alpha$

$\Rightarrow 1 + \hat{y}_t = (1 + \hat{a}_t) \cdot (1 + \alpha \hat{k}_t) (1 + (1-\alpha) \hat{n}_t)$ $g(0) = 1, g'(0) = \alpha$

$\Rightarrow 1 + \hat{y}_t = (1 + \hat{a}_t + \alpha \hat{k}_t + (1-\alpha) \hat{n}_t)$

$\Rightarrow 1 + \hat{y}_t = 1 + \hat{a}_t + \alpha \hat{k}_t + (1-\alpha) \hat{n}_t + \underbrace{\frac{(1-\alpha)\hat{n}_t\hat{a}_t}{2}}_{\approx 0} + \underbrace{\frac{(1-\alpha)\alpha\hat{n}_t\hat{k}_t}{2}}_{\approx 0} \Rightarrow \hat{y}_t = \hat{a}_t + \alpha \hat{k}_t + (1-\alpha) \hat{n}_t$

⑨ GDP identity: $Y_t = C_t + I_t + G_t$

From ⑥ we know $\Rightarrow \hat{y}_t = \frac{\bar{C}}{\bar{Y}} \hat{c}_t + \frac{\bar{I}}{\bar{Y}} \hat{i}_t + \frac{\bar{G}}{\bar{Y}} \hat{g}_t = s_C \hat{c}_t + s_I \hat{i}_t + s_G \hat{g}_t$ share of GDP for each component

(10) Capital law of motion: $K_{t+1} = (1-\delta)K_t + I_t \Rightarrow \bar{K} = (1-\delta)\bar{K} + \bar{I}$

$$\Rightarrow \bar{K}(1 + \hat{k}_{t+1}) = (1-\delta)\bar{K}(1 + \hat{k}_t) + \bar{I}(1 + \hat{i}_t)$$

$$\Rightarrow \cancel{\bar{K}} + \bar{K}\hat{k}_{t+1} = \cancel{(1-\delta)\bar{K}} + (1-\delta)\bar{K}\hat{k}_t + \cancel{\bar{I}} + \bar{I}\hat{i}_t$$

$$\Rightarrow \bar{K}\hat{k}_{t+1} = (1-\delta)\bar{K}\hat{k}_t + \bar{I}\hat{i}_t$$

$$\Rightarrow \hat{k}_{t+1} = (1-\delta)\hat{k}_t + \frac{\bar{I}}{\bar{K}}\hat{i}_t \quad \Big| \Rightarrow \hat{k}_{t+1} = (1-\delta)\hat{k}_t + \delta\hat{i}_t$$

$$\bar{K} = (1-\delta)\bar{K} + \bar{I} \Rightarrow \frac{\bar{I}}{\bar{K}} = \delta$$

(11) Euler equation: $u'(C_t) = \beta E_t [u'(C_{t+1}) R_{t+1}]$

use CRRA utility: $u(C_t) = \frac{C_t^{1-\bar{\sigma}}}{1-\bar{\sigma}} \Rightarrow u'(C_t) = C_t^{-\bar{\sigma}}$

$$\Rightarrow C_t^{-\bar{\sigma}} = \beta E_t [C_{t+1}^{-\bar{\sigma}} R_{t+1}] \Rightarrow \bar{C}^{-\bar{\sigma}} = \beta \bar{C}^{-\bar{\sigma}} \bar{R} \Rightarrow 1 = \beta \bar{R}$$

let $C_t = \bar{C} e^{\hat{C}_t}$, $C_{t+1} = \bar{C} e^{\hat{C}_{t+1}}$, $R_{t+1} = \bar{R} e^{\hat{R}_{t+1}}$

$$\Rightarrow \bar{C}^{-\bar{\sigma}} e^{-\bar{\sigma}\hat{C}_t} = \beta E_t [\bar{C}^{-\bar{\sigma}} e^{-\bar{\sigma}\hat{C}_{t+1}} \cdot \bar{R} e^{\hat{R}_{t+1}}]$$

$$\Rightarrow \bar{C}^{-\bar{\sigma}} (1 - \bar{\sigma}\hat{C}_t) = \beta E_t [\bar{C}^{-\bar{\sigma}} (1 - \bar{\sigma}\hat{C}_{t+1}) (1 + \hat{R}_{t+1})]$$

$$\Rightarrow \bar{C}^{-\bar{\sigma}} - \bar{C}^{-\bar{\sigma}} \bar{\sigma}\hat{C}_t = \beta E_t [\bar{C}^{-\bar{\sigma}} \bar{R} (1 + \hat{R}_{t+1} - \bar{\sigma}\hat{C}_{t+1} - \underbrace{|\bar{\sigma}\hat{C}_{t+1}\hat{R}_{t+1}|}_{\approx 0})]$$

$$\Rightarrow \bar{C}^{-\bar{\sigma}} - \bar{C}^{-\bar{\sigma}} \bar{\sigma}\hat{C}_t = \underbrace{\beta \bar{R}}_{=1} \bar{C}^{-\bar{\sigma}} [1 + E_t (-\bar{\sigma}\hat{C}_{t+1} + \hat{R}_{t+1})]$$

$$\Rightarrow 1 - \bar{\sigma}\hat{C}_t = 1 + E_t (-\bar{\sigma}\hat{C}_{t+1} + \hat{R}_{t+1})$$

$$\Rightarrow -\bar{\sigma}\hat{C}_t = E_t (-\bar{\sigma}\hat{C}_{t+1} + \hat{R}_{t+1})$$

$$\Rightarrow \hat{C}_t = E_t (\hat{C}_{t+1}) - \frac{1}{\bar{\sigma}} E_t (\hat{R}_{t+1})$$

★ Log linearization cheat sheet for economics.

① XY : $\hat{x} + \hat{y}$

② X/Y : $\hat{x} - \hat{y}$

③ X^α : $\alpha \hat{x}$

④ $\prod_i X_i^{\alpha_i}$: $\sum_i \alpha_i \hat{x}_i$

⑤ $\sum_i X_i$: $\sum_i \frac{\bar{X}_i}{\sum_j \bar{X}_j} \hat{x}_i$

⑥ $K' = (1-\delta)K + I$: $\hat{k}' = (1-\delta)\hat{k} + \delta\hat{i}$

⑦ $\pi = \frac{p}{p-1}$: $\hat{\pi} = \hat{p} - \hat{p}_{-1}$

⑧ CRRA MU: $\hat{u}'(C_t) = -\bar{\sigma}\hat{C}_t$

⑨ CES index $X = [\sum_i w_i X_i^\rho]^{1/\rho}$, $\hat{x} = \sum_i s_i \hat{x}_i$

with $s_i = \frac{w_i \bar{X}_i^\rho}{\sum_j w_j \bar{X}_j^\rho}$